

grow faster than the portfolio balance, and decreasing real interest rates increase the present value of the spending stream. Since the value of Treasury bills does not grow sufficiently when interest rates drop, sustainable spending falls. Though Treasury bills can keep nominal wealth stable, the real spending they can support is actually quite volatile.

Contrast that with a portfolio of TIPS with the same duration as the spending goal. TIPS provide inflation protection, so if inflation rises, they support greater income to match what is needed for the real spending goal. When the TIPS portfolio has the same duration as the spending objective, interest rate fluctuations are also hedged. When interest rates decrease, the cost of the spending objective increases, and the value of the TIPS portfolio rises accordingly. If interest rates rise, then the TIPS portfolio loses value, but the cost of meeting the spending objective also decreases.

For a portfolio of long-term TIPS, the account balances may be quite volatile as inflation and interest rates fluctuate, but the sustainable amount of inflation-adjusted spending remains reasonably level. Exhibit 7.7 illustrates these ideas: Treasury bills support low *wealth* volatility but high *spending* volatility, while TIPS support low spending volatility but high wealth volatility. TIPS users accept price risk to meet the spending goal.

*Exhibit 7.7*

*Trade-Offs in Retirement*